Averaged Stochastic Loss-Surface Analysis of the Hybrid Model

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Abstract

This report investigates the geometry of the stochastic loss surface of the Hybrid behavioural model after averaging over multiple simulation seeds. In the previous analysis, local loss-surface slices were computed using a single stochastic evaluation per point. Here, each parameter setting was evaluated across 10 independent stochastic seeds, and the mean loss together with the standard error of the mean (SEM) were analysed. Random one-dimensional slices through the seven-dimensional parameter space were explored over a substantially larger displacement range. Surprisingly, even after averaging across stochastic simulations, large lower-loss regions remained accessible near the fitted solutions. In several cases, the lower-loss points occurred far from the fitted parameter vectors and frequently interacted with optimisation boundaries. These findings suggest that the Hybrid-model objective landscape contains substantial geometric complexity, potentially including broad valleys, plateaus, saddle-like regions, anisotropic curvature, or weak parameter identifiability.

Keywords

Hybrid model, stochastic optimisation, loss surface, CMA-ES, TPE, behavioural modelling

1. Methods

The aim of this analysis was to investigate the geometry of the averaged stochastic loss landscape around fitted parameter values of the Hybrid model. The fitted parameter vector was defined as

$$\boldsymbol{\theta}^* = \left[\sigma_{\text{noise}}, A_{\text{repulsion}}, \gamma, \sigma_{\text{update}}, \eta_{\text{learning}}, \sigma_{\text{boundary}}, \alpha\right]^\top \in \mathbb{R}^7. \tag{1}$$

The seven fitted parameters were constrained to the following optimisation ranges:

$$\sigma_{\text{noise}} \in [0.01, 0.45], \tag{2}$$
$$A_{\text{repulsion}} \in [0, 0.5], \tag{3}$$
$$\gamma \in [0.01, 1], \tag{4}$$
$$\sigma_{\text{update}} \in [0.01, 0.7], \tag{5}$$
$$\eta_{\text{learning}} \in [0.01, 1], \tag{6}$$
$$\sigma_{\text{boundary}} \in [0.01, 10], \tag{7}$$
$$\alpha \in [0, 1]. \tag{8}$$

For a parameter vector $\boldsymbol{\theta}$ and stochastic seed s, the loss function was denoted by

$$\mathcal{L}(\boldsymbol{\theta}; s). \tag{9}$$

To reduce stochastic fluctuations in the estimated loss surface, each parameter setting was evaluated across 10 independent stochastic seeds:

$$\mathcal{S} = \{1, 2, \dots, 10\}. \tag{10}$$

The averaged stochastic loss was then defined as

$$\bar{\mathcal{L}}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \mathcal{L}(\boldsymbol{\theta}; s). \tag{11}$$

The uncertainty across seeds was quantified using the standard error of the mean:

$$\mathrm{SEM}(\boldsymbol{\theta}) = \frac{\mathrm{std}\left(\{\mathcal{L}(\boldsymbol{\theta}; s)\}_{s \in \mathcal{S}}\right)}{\sqrt{|\mathcal{S}|}}. \tag{12}$$

To probe the local geometry of the loss landscape, random directions were sampled in the seven-dimensional parameter space. A raw random vector was first sampled from an isotropic Gaussian distribution:

$$\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I_7). \tag{13}$$

Because the parameters have very different numerical scales, the sampled vector was scaled by the width of each parameter range:

$$r_i = \theta_{i,\max} - \theta_{i,\min}. \tag{14}$$

The scaled direction vector was then normalised:

$$\mathbf{v} = \frac{\mathbf{z} \odot \mathbf{r}}{\|\mathbf{z} \odot \mathbf{r}\|_2}, \tag{15}$$

where $\odot$ denotes element-wise multiplication.

For each sampled direction $\mathbf{v}$, a one-dimensional slice through the fitted point was explored:

$$\boldsymbol{\theta}(\lambda) = \mathrm{clip}\left(\boldsymbol{\theta}^* + \lambda \mathbf{v}\right), \tag{16}$$

where $\mathrm{clip}(\cdot)$ enforces the optimisation bounds.

The displacement parameter was explored over the interval

$$\lambda \in [-5, 5]. \tag{17}$$

The interval was sampled using 201 evenly spaced values:

$$\Delta\lambda = \frac{10}{200} = 0.05. \tag{18}$$

Since the sampled direction vectors were normalised to unit norm, the displacement parameter λ approximately corresponds to Euclidean distance travelled in parameter space, except when clipping at parameter boundaries occurs.

For each subject and optimisation method, $D = 10$ random directions were sampled. Therefore, each fitted solution was evaluated at

$$10 \times 201 = 2010 \tag{19}$$

parameter settings, each of which was averaged across 10 stochastic simulations.

2. Analysis

2.1. Subject 6269757 fitted with CMA-ES

For subject 6269757 fitted using CMA-ES, the averaged fitted loss across seeds was

$$\bar{\mathcal{L}}(\boldsymbol{\theta}^*) = 0.133142 \tag{20}$$

with

$$\text{SEM} = 0.012003. \tag{21}$$

The fitted parameter vector was

$$\begin{aligned} \sigma_{\text{noise}} &= 0.118160, \\ A_{\text{repulsion}} &= 0.264088, \\ \gamma &= 0.842005, \\ \sigma_{\text{update}} &= 0.486071, \\ \eta_{\text{learning}} &= 0.336001, \\ \sigma_{\text{boundary}} &= 4.726866, \\ \alpha &= 0.693673. \end{aligned} \tag{22}$$

The best point discovered during the random-direction exploration had averaged loss

$$\bar{\mathcal{L}}_{\text{best}} = 0.110205, \tag{23}$$

corresponding to an improvement of approximately

$$\frac{0.133142 - 0.110205}{0.133142} \times 100 \approx 17.23\%. \tag{24}$$

The best-found parameter vector was

$$\begin{aligned} \sigma_{\text{noise}} &= 0.172829, \\ A_{\text{repulsion}} &= 0.186923, \\ \gamma &= 0.831871, \\ \sigma_{\text{update}} &= 0.538076, \\ \eta_{\text{learning}} &= 0.535494, \\ \sigma_{\text{boundary}} &= 5.389020, \\ \alpha &= 0.691575. \end{aligned} \tag{25}$$

The best point occurred in direction 2 at

$$\lambda = -0.70, \tag{26}$$

with Euclidean distance

$$d \approx 0.70. \tag{27}$$

Interestingly, despite the relatively large displacement and substantial changes in several parameters, the hybrid mixing parameter α remained nearly unchanged:

$$\alpha_{\text{fitted}} = 0.693673, \tag{28}$$

while

$$\alpha_{\text{best}} = 0.691575. \tag{29}$$

2.2. Subject 6269757 fitted with TPE

For subject 6269757 fitted using TPE, the averaged fitted loss was

$$\bar{\mathcal{L}}(\boldsymbol{\theta}^{*}) = 0.123259 \tag{30}$$

with

$$\text{SEM} = 0.010905. \tag{31}$$

The fitted parameter vector was

$$\begin{aligned} \sigma_{\text{noise}} &= 0.126412, \\ A_{\text{repulsion}} &= 0.339304, \\ \gamma &= 0.939783, \\ \sigma_{\text{update}} &= 0.272976, \\ \eta_{\text{learning}} &= 0.547309, \\ \sigma_{\text{boundary}} &= 9.295416, \\ \alpha &= 0.451329. \end{aligned} \tag{32}$$

The best discovered point had averaged loss

$$\bar{\mathcal{L}}_{\text{best}} = 0.112922, \tag{33}$$

corresponding to an improvement of approximately

$$\frac{0.123259 - 0.112922}{0.123259} \times 100 \approx 8.39\%. \tag{34}$$

The best-found parameter vector was

$$\begin{aligned} \sigma_{\text{noise}} &= 0.150028, \\ A_{\text{repulsion}} &= 0.334467, \\ \gamma &= 0.962180, \\ \sigma_{\text{update}} &= 0.265100, \\ \eta_{\text{learning}} &= 0.510864, \\ \sigma_{\text{boundary}} &= 10.000000, \\ \alpha &= 0.438998. \end{aligned} \tag{35}$$

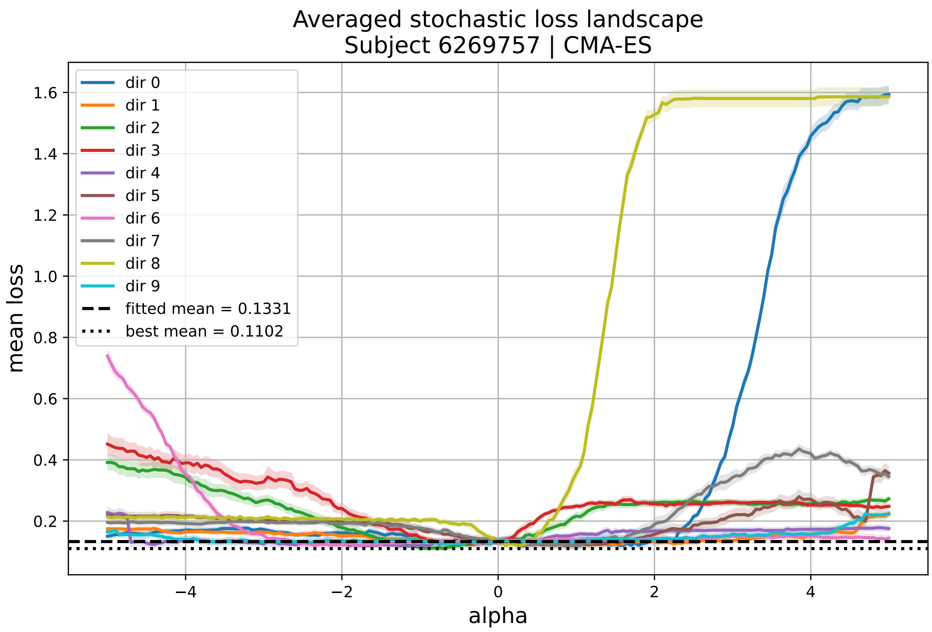

Figure 1: Averaged stochastic loss surface for subject 6269757 fitted with CMA-ES. Each curve represents the mean loss across 10 stochastic seeds and shaded regions indicate the SEM.

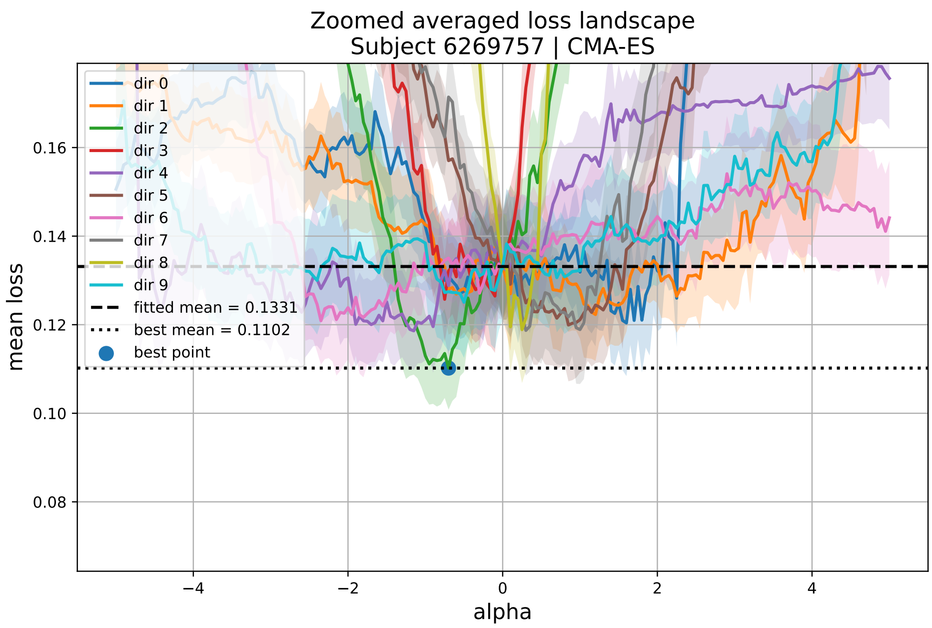

Figure 2: Zoomed averaged stochastic loss surface for subject 6269757 fitted with CMA-ES. The lower-loss region persists after averaging over stochastic simulations.

The best point occurred in direction 9 at

$$\lambda = -0.80, \tag{36}$$

with Euclidean distance

$$d \approx 0.706. \tag{37}$$

Importantly, the best-found point pushed

$$\sigma_{\text{boundary}} \tag{38}$$

to the upper optimisation boundary:

$$\sigma_{\text{boundary}} = 10. \tag{39}$$

2.3. Subject 6269770 fitted with CMA-ES

For subject 6269770 fitted using CMA-ES, the averaged fitted loss was

$$\bar{\mathscr{L}}(\boldsymbol{\theta}^*) = 0.044032 \tag{40}$$

with

$$\text{SEM} = 0.002789. \tag{41}$$

The fitted parameter vector was

$$\begin{aligned} \sigma_{\text{noise}} &= 0.104845, \\ A_{\text{repulsion}} &= 0.459508, \\ \gamma &= 0.770726, \\ \sigma_{\text{update}} &= 0.398532, \\ \eta_{\text{learning}} &= 0.688318, \\ \sigma_{\text{boundary}} &= 0.388483, \\ \alpha &= 0.979646. \end{aligned} \tag{42}$$

The best-found point had averaged loss

$$\bar{\mathscr{L}}_{\text{best}} = 0.034169, \tag{43}$$

corresponding to an improvement of approximately

$$\frac{0.044032 - 0.034169}{0.044032} \times 100 \approx 22.40\%. \tag{44}$$

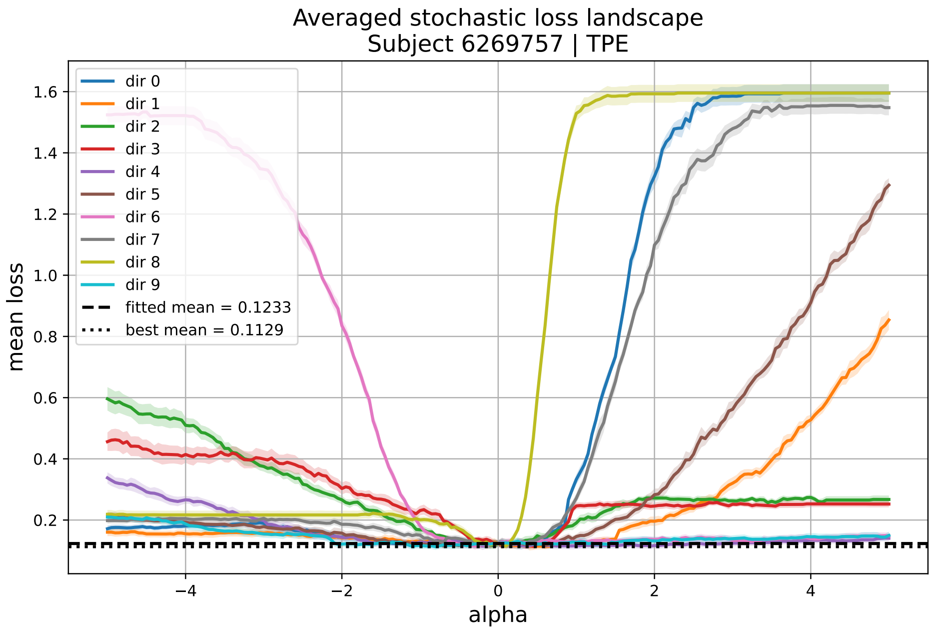

Figure 3: Averaged stochastic loss surface for subject 6269757 fitted with TPE. Lower-loss regions remain accessible after stochastic averaging.

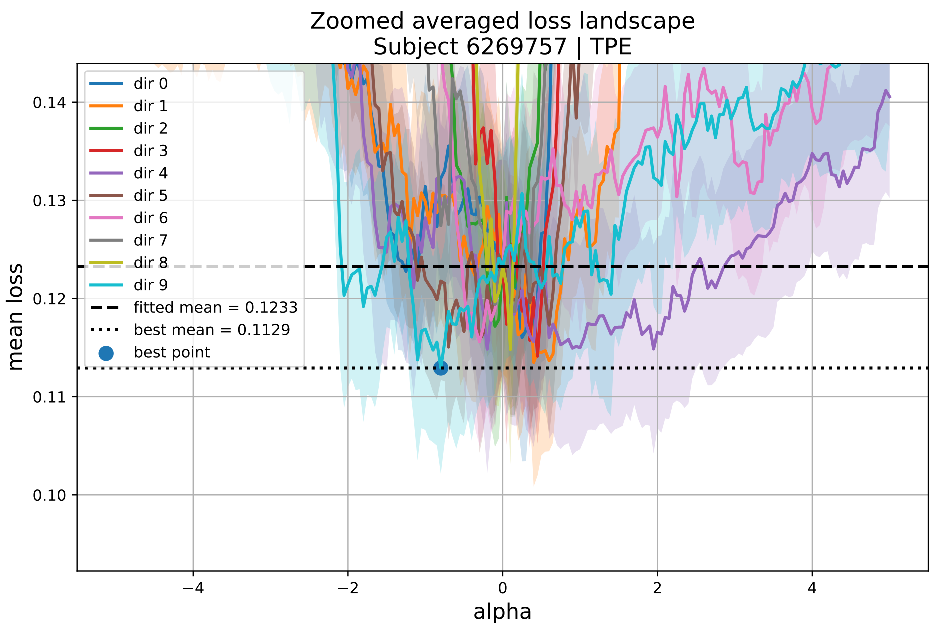

Figure 4: Zoomed averaged stochastic loss surface for subject 6269757 fitted with TPE. The best point occurs near the upper optimisation boundary of σ_{boundary}.

The best-found parameter vector was

$$\begin{aligned} \sigma_{\text{noise}} &= 0.120145, \\ A_{\text{repulsion}} &= 0.446031, \\ \gamma &= 0.685166, \\ \sigma_{\text{update}} &= 0.404591, \\ \eta_{\text{learning}} &= 0.693977, \\ \sigma_{\text{boundary}} &= 0.010000, \\ \alpha &= 0.943818. \end{aligned} \tag{45}$$

The best point occurred in direction 5 at

$$\lambda = 0.65. \tag{46}$$

Interestingly, the parameter

$$\sigma_{\text{boundary}} \tag{47}$$

collapsed to the lower optimisation boundary:

$$\sigma_{\text{boundary}} = 0.01. \tag{48}$$

2.4. Subject 6269770 fitted with TPE

For subject 6269770 fitted using TPE, the averaged fitted loss was

$$\bar{\mathscr{L}}(\boldsymbol{\theta}^*) = 0.043197 \tag{49}$$

with

$$\text{SEM} = 0.004870. \tag{50}$$

The fitted parameter vector was

$$\begin{aligned} \sigma_{\text{noise}} &= 0.099011, \\ A_{\text{repulsion}} &= 0.499931, \\ \gamma &= 0.415736, \\ \sigma_{\text{update}} &= 0.204130, \\ \eta_{\text{learning}} &= 0.310139, \\ \sigma_{\text{boundary}} &= 6.621977, \\ \alpha &= 0.956196. \end{aligned} \tag{51}$$

The best-found point had averaged loss

$$\bar{\mathscr{L}}_{\text{best}} = 0.032240, \tag{52}$$

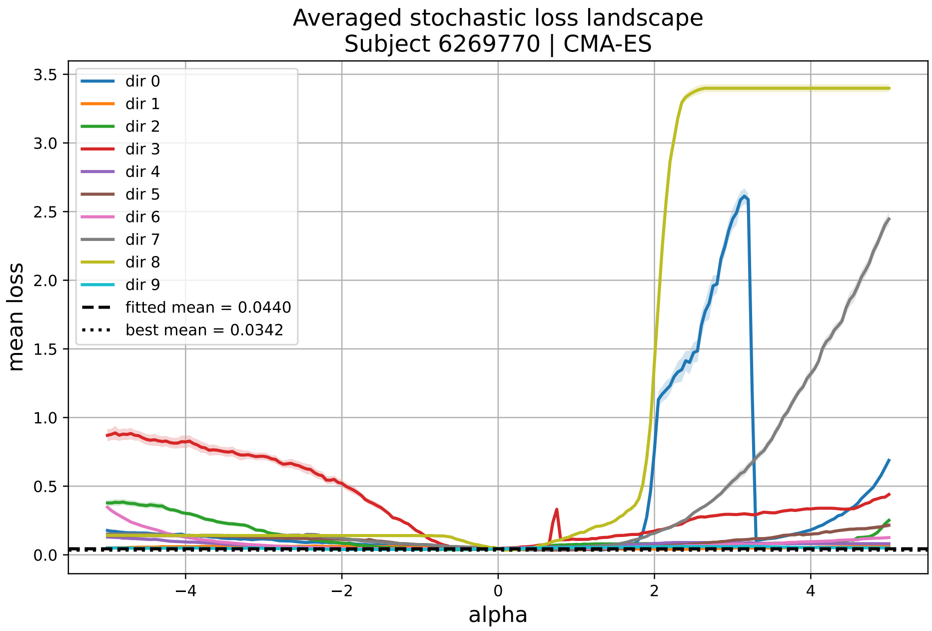

Figure 5: Averaged stochastic loss surface for subject 6269770 fitted with CMA-ES. Large lower-loss regions remain visible after averaging across stochastic seeds.

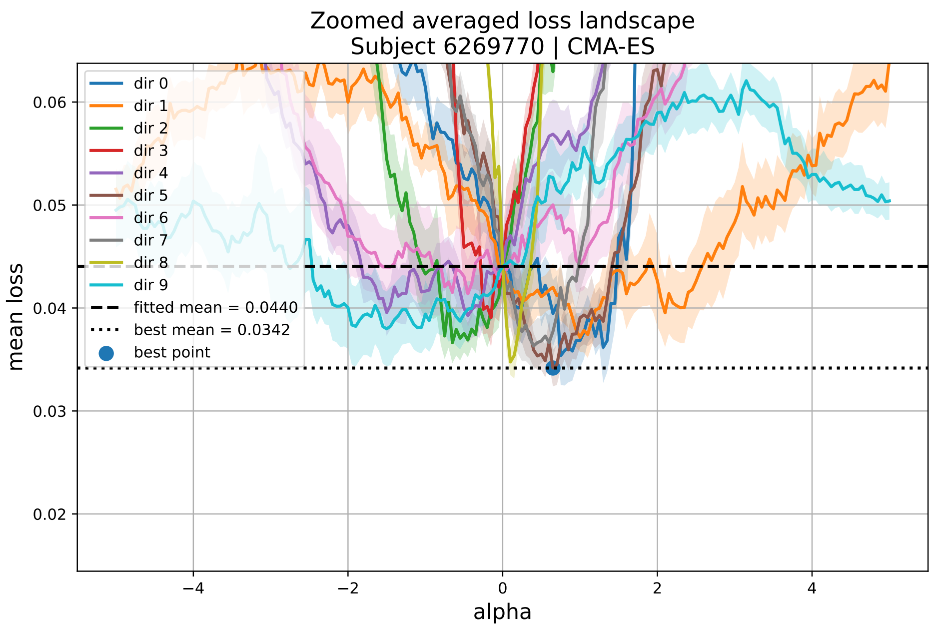

Figure 6: Zoomed averaged stochastic loss surface for subject 6269770 fitted with CMA-ES. The best point occurs near the lower optimisation boundary of σ_{boundary}.

corresponding to an improvement of approximately

$$\frac{0.043197 - 0.032240}{0.043197} \times 100 \approx 25.36\%. \tag{53}$$

The best-found parameter vector was

$$\begin{aligned} \sigma_{\text{noise}} &= 0.117909, \\ A_{\text{repulsion}} &= 0.500000, \\ \gamma &= 0.473735, \\ \sigma_{\text{update}} &= 0.164852, \\ \eta_{\text{learning}} &= 0.289136, \\ \sigma_{\text{boundary}} &= 9.637303, \\ \alpha &= 0.602223. \end{aligned} \tag{54}$$

The best point was found in direction 6 at

$$\lambda = -3.05, \tag{55}$$

with Euclidean distance

$$d \approx 3.04. \tag{56}$$

This was by far the largest displacement observed in the analysis. Unlike the previous examples, the hybrid mixing parameter changed substantially:

$$\alpha_{\text{fitted}} = 0.956196, \tag{57}$$

while

$$\alpha_{\text{best}} = 0.602223. \tag{58}$$

At the same time,

$$\sigma_{\text{boundary}} \tag{59}$$

increased strongly:

$$\sigma_{\text{boundary,fitted}} = 6.621977, \tag{60}$$

while

$$\sigma_{\text{boundary,best}} = 9.637303. \tag{61}$$

and the parameter

$$A_{\text{repulsion}} \tag{62}$$

collapsed to the upper optimisation boundary:

$$A_{\text{repulsion}} = 0.50. \tag{63}$$

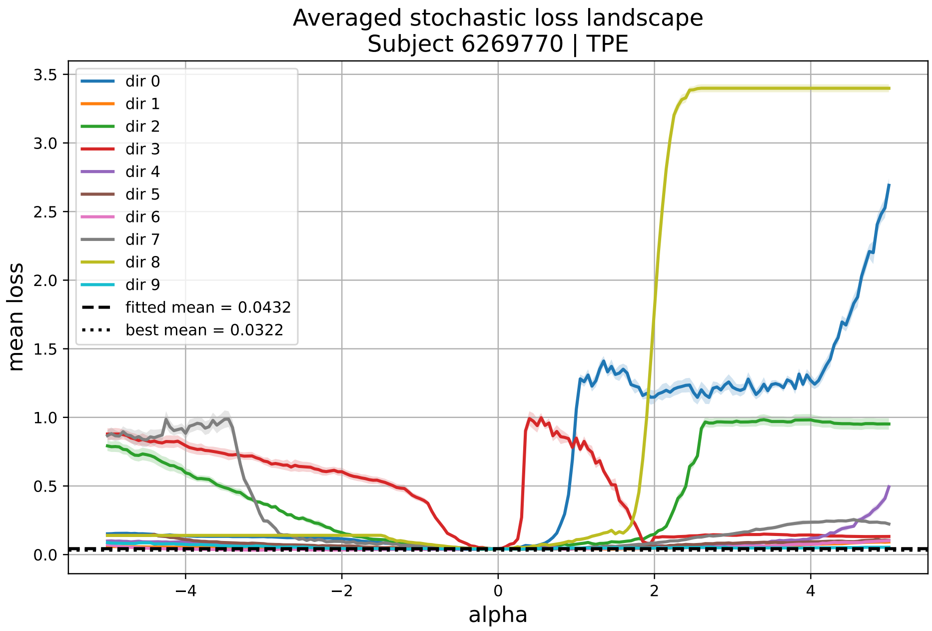

Figure 7: Averaged stochastic loss surface for subject 6269770 fitted with TPE. Lower-loss regions remain visible even after averaging across 10 stochastic seeds.

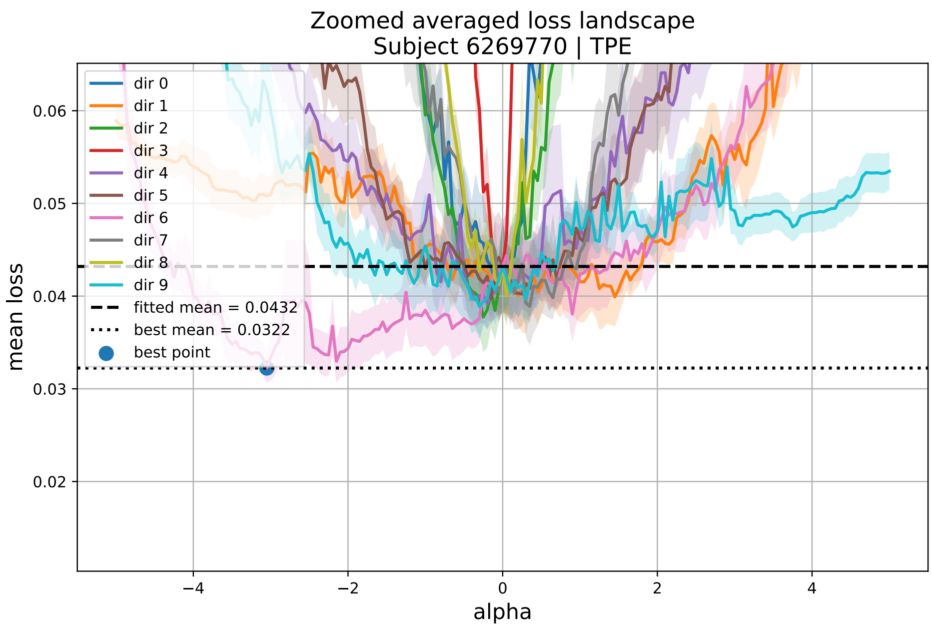

Figure 8: Zoomed averaged stochastic loss surface for subject 6269770 fitted with TPE. The best point occurs at a large displacement from the fitted solution and involves major changes in both α and σ_{boundary}.

2.5. Summary of numerical results

2.6. Interpretation

The averaged stochastic analysis strengthens the conclusions of the previous single-seed investigation. Even after averaging each parameter setting across 10 independent stochastic simulations and visualising SEM uncertainty bands, large lower-loss regions remained consistently accessible around the fitted solutions.

This persistence strongly suggests that the observed improvements are not merely caused by stochastic fluctuations of the simulation process. Instead, they appear to reflect stable geometric properties of the underlying objective landscape.

Several striking behaviours emerged repeatedly throughout the analysis. First, lower-loss regions were often found relatively far from the fitted parameter vectors. In particular, the TPE solution for subject 6269770 exhibited a substantially lower-loss point at Euclidean distance greater than 3 from the fitted solution, together with a major change in the hybrid mixing parameter α. This suggests that at least some fitted solutions may not correspond to robust local optima of the averaged stochastic objective.

Second, several lower-loss solutions strongly interacted with optimisation boundaries. In some cases, σ_{boundary} collapsed to its minimum allowed value, while in other cases it saturated at the maximum allowed value. This repeated boundary-hitting behaviour suggests that the imposed optimisation constraints may be actively shaping the geometry of the explored landscape, or alternatively that the true optimum may lie partially outside the explored parameter region.

Third, some cases showed large improvements without substantial changes in α, while others involved dramatic shifts in α. This suggests that the geometry of the loss landscape may vary considerably across subjects and optimisation methods, with different parameter subspaces contributing differently to the optimisation dynamics.

Overall, these results point toward substantial geometric complexity of the Hybrid-model objective function. Possible explanations include broad plateaus, elongated valleys, saddle-like regions, weak parameter identifiability, anisotropic curvature across dimensions, or interactions between stochastic simulation noise and optimisation boundaries. The results also suggest that fitted solutions obtained from standard optimisation procedures should be interpreted cautiously, since substantial lower-loss regions may remain accessible even after stochastic averaging.

A useful next step would be to use the newly identified lower-loss points as initialisations for additional optimisation stages, or to apply adaptive local search procedures around the fitted solutions in order to better characterise the nearby geometry of the objective landscape.

Table 1
Summary of averaged stochastic loss-surface analysis results.

Subject	Optimiser	Mean fitted loss	Best mean loss	Improvement (%)	Direction	Best λ
6269757	CMA-ES	0.133142	0.110205	17.23	2	-0.70
6269757	TPE	0.123259	0.112922	8.39	9	-0.80
6269770	CMA-ES	0.044032	0.034169	22.40	5	0.65
6269770	TPE	0.043197	0.032240	25.36	6	-3.05